



UNSW
SYDNEY

MATH2601 – Higher Linear Algebra

Topic 2 – Vector spaces

Lecture 2.1 – Vector spaces and subspaces

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Vector spaces

Recall the following definition of a vector space from MATH1241.

Definition. A **vector space** is a set V (of **vectors**) equipped with a field $\mathbb{F}$ (of **scalars**) and two binary operators **vector addition** $(+)$ and **scalar multiplication** $(\cdot)$ or concatenation) satisfying the following axioms:

- **Closure under addition:** $\mathbf{u} + \mathbf{v} \in V$ for all $\mathbf{u}, \mathbf{v} \in V$.
- **Additive associativity:** $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$.
- **Additive commutativity:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v} \in V$.
- **Additive identity:** There exists $\mathbf{0} \in V$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v} \in V$.
- **Additive inverse:** For all $\mathbf{v} \in V$ there is $-\mathbf{v} \in V$ such that $\mathbf{v} + -\mathbf{v} = \mathbf{0}$.
- **Closure under scalar multiplication:** $\lambda \mathbf{v} \in V$ for all $\lambda \in \mathbb{F}$ and $\mathbf{v} \in V$.
- **Scalar multiplicative associativity:** $(\lambda \mu) \mathbf{v} = \lambda (\mu \mathbf{v})$ for all $\lambda, \mu \in \mathbb{F}$, $\mathbf{v} \in V$.
- **Scalar multiplicative identity:** $1 \mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in V$,
where 1 is the multiplicative identity in $\mathbb{F}$.
- **Vector distributivity:** $\lambda (\mathbf{u} + \mathbf{v}) = \lambda \mathbf{u} + \lambda \mathbf{v}$ for all $\lambda \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in V$.
- **Scalar distributivity:** $(\lambda + \mu) \mathbf{v} = \lambda \mathbf{v} + \mu \mathbf{v}$ for all $\lambda, \mu \in \mathbb{F}$ and $\mathbf{v} \in V$.

Vector spaces in terms of groups

We can now slightly refine our definition of a vector space:

Definition. A **vector space** is a set V (of **vectors**) equipped with a field $\mathbb{F}$ (of **scalars**) and two binary operators **vector addition** $(+)$ and **scalar multiplication** $(\cdot)$ or concatenation) satisfying the following axioms:

- $(V, +)$ is an abelian group.
- $\lambda \mathbf{v} \in V$ for all $\lambda \in \mathbb{F}$ and $\mathbf{v} \in V$. (**Closure under $\cdot$.**)
- $(\lambda \mu) \mathbf{v} = \lambda(\mu \mathbf{v})$ for all $\lambda, \mu \in \mathbb{F}$, $\mathbf{v} \in V$. (**Associativity of $\cdot$.**)
- $1 \mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in V$, where 1 is the identity of $(\mathbb{F}, \times)$. (**Identity for $\cdot$.**)
- $\lambda(\mathbf{u} + \mathbf{v}) = \lambda \mathbf{u} + \lambda \mathbf{v}$ for all $\lambda \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in V$. (**Vector distributivity.**)
- $(\lambda + \mu) \mathbf{v} = \lambda \mathbf{v} + \mu \mathbf{v}$ for all $\lambda, \mu \in \mathbb{F}$ and $\mathbf{v} \in V$. (**Scalar distributivity.**)

Notation. Since the definition of a vector space is in terms of the set V , the field $\mathbb{F}$, and the operations $+$ and $\cdot$, a vector space can be written as the tuple $(V, \mathbb{F}, +, \cdot)$, the pair $(V, \mathbb{F})$, or just the vector set V .

Note that scalar multiplication is a function $\cdot : \mathbb{F} \times V \rightarrow V$.

We typically use boldface for vector elements to distinguish them from scalar elements. In particular, the additive identity in V is $\mathbf{0}$, as opposed to 0 in $\mathbb{F}$.

Matrices and column vectors

Example. The **trivial vector space** is $\{\mathbf{0}\}$ (over any scalar field).

Example. Any **field** $\mathbb{F}$ is a vector space over itself.

Example. The set $M_{p,q}(\mathbb{F})$ of $p \times q$ **matrices** with entries from $\mathbb{F}$ is a vector space over $\mathbb{F}$ with vector addition defined as matrix addition, and scalar multiplication defined as multiplying each entry of the matrix by the scalar.

Example. The set $\mathbb{F}^n = M_{n,1}(\mathbb{F})$ of **column vectors** is a vector space over $\mathbb{F}$. This is an especially common type of vector space, often denoted $\mathbb{F}^n$ instead of $(\mathbb{F}^n, \mathbb{F}, +, *)$.

Recall that for a particular column vector $\mathbf{v} \in \mathbb{F}^n$, we write

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \quad \text{where each } v_i \in \mathbb{F}$$

or just $\mathbf{v} = (v_i)_{1 \leq i \leq n}$, and refer to v_i as the **i th component** or entry of $\mathbf{v}$.

In particular, we can write that given $\mathbf{v}, \mathbf{u} \in \mathbb{F}^n$, vector addition in $\mathbb{F}^n$ is given by $\mathbf{u} + \mathbf{v} = (u_i + v_i)_{1 \leq i \leq n}$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{F}^n$, and scalar multiplication in $\mathbb{F}^n$ is given by $\lambda \mathbf{v} = (\lambda v_i)_{1 \leq i \leq n}$ for all $\lambda \in \mathbb{F}$ and $\mathbf{v} \in \mathbb{F}^n$.

Examples of vector spaces

Example. Which of the following sets are vector spaces?

- $(\mathbb{C}^5, \mathbb{C}, +, *)$ **is** a vector space, namely $\mathbb{C}^5$.
- $(\mathbb{R}^2, \mathbb{C}, +, *)$ is **not** a vector space, since it is not closed under scalar multiplication. For example, choosing $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \in \mathbb{R}^2$ and $\lambda = i \in \mathbb{C}$,

we have $\lambda \mathbf{v} = \begin{pmatrix} i \\ i \end{pmatrix} \notin \mathbb{R}^2$.

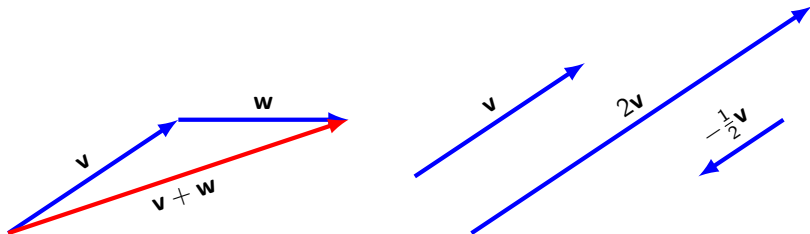
- $(\mathbb{C}^4, \mathbb{R}, +, *)$ **is** a vector space, since it is a set of complex column vectors equipped with the usual operations and a scalar field that is a subfield of the usual scalar field $\mathbb{C}$.
- Recalling that $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$,
 $(\mathbb{Q}(\sqrt{2}), \mathbb{Q}, +, *)$ **is** a vector space, since $\mathbb{Q}(\sqrt{2})$ is a field and $\mathbb{Q}$ is a subfield of $\mathbb{Q}(\sqrt{2})$.

Alternatively, note that $\phi : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}(\sqrt{2})$ given by $\phi(a, b) = a + b\sqrt{2}$ for all $a, b \in \mathbb{Q}$ is an isomorphism (prove this!), and $\mathbb{Q} \times \mathbb{Q} \cong \mathbb{Q}^2$ is a known vector space over $\mathbb{Q}$.

Geometric vectors

Definition. A **geometric vector** (or **Euclidean vector**) in $\mathbb{R}^n$ (for any $n \in \mathbb{Z}^+$) is characterised as a geometric object with both magnitude and direction. It is identified as an ordered pair of points in $\mathbb{R}^n$, represented visually as an arrow stretching from the first point to the second. We treat two geometric vectors as **equivalent** if one is a translation of the other.

Example. The set of geometric vectors in $\mathbb{R}^n$ is a vector space over $\mathbb{R}$, with vector addition defined by taking the displacement after joining the vectors head-to-tail and scalar multiplication defined by stretching the vector by a factor of the scalar multiple (reversing direction if negative).



Function spaces

Definition. For any field $\mathbb{F}$ and nonempty set X , the set $\mathcal{F}(X)$ (or $\mathbb{F}^X$) is the set of all functions from X to $\mathbb{F}$, that is, $\mathcal{F}(X) = \{f : X \rightarrow \mathbb{F}\}$.

Example. The set $\mathcal{F}(X)$ is a vector space (called the **function space**) over $\mathbb{F}$, with vector addition defined for all $f, g \in \mathcal{F}(X)$ by

$$(f + g)(x) = f(x) + g(x) \text{ for all } x \in X,$$

and scalar multiplication defined for all $f \in \mathcal{F}(X)$ and $\lambda \in \mathbb{F}$ by

$$(\lambda f)(x) = \lambda(f(x)) \text{ for all } x \in X.$$

Note that the additive identity in $\mathcal{F}(X)$ is the **zero function** $z : X \rightarrow \mathbb{F}$ given by $z(x) = 0$ for all $x \in X$.

Definition. For any field $\mathbb{F}$, the set of all **polynomials over $\mathbb{F}$** is denoted $\mathcal{P}(\mathbb{F})$ and is considered a subset of $\mathcal{F}(\mathbb{F})$. It contains all (finite-degree) polynomials with coefficients in $\mathbb{F}$.

For any $n \in \mathbb{N}$, the set of all **polynomials over $\mathbb{F}$ of degree at most n** is denoted $\mathcal{P}_n(\mathbb{F})$. Note that the degree of the zero polynomial is not well-defined, but for our purposes we will treat it as $-\infty$.

Example. $\mathcal{P}(\mathbb{F})$ and $\mathcal{P}_n(\mathbb{F})$ are vector spaces over $\mathbb{F}$, for any field $\mathbb{F}$.

Non-standard example

Exercise. Show that $(\mathbb{R}^+, \mathbb{R}, \oplus, \otimes)$ is a vector space, where $\mathbf{u} \oplus \mathbf{v} = \mathbf{uv}$ and $\lambda \otimes \mathbf{v} = \mathbf{v}^\lambda$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^+$ and $\lambda \in \mathbb{R}$.

Solution. First, notice that $(\mathbb{R}^+, \oplus) = (\mathbb{R}^+, \times)$ is an abelian group. It remains to check the scalar multiplication axioms.

Closure: For any $\mathbf{v} \in \mathbb{R}^+$ and $\lambda \in \mathbb{R}$, we have $\lambda \otimes \mathbf{v} = \mathbf{v}^\lambda \in \mathbb{R}^+$.

Associativity: For any $\mathbf{v} \in \mathbb{R}^+$ and $\lambda, \mu \in \mathbb{R}$, we have

$$\lambda \otimes (\mu \otimes \mathbf{v}) = (\mathbf{v}^\mu)^\lambda = \mathbf{v}^{\lambda\mu} = (\lambda\mu) \otimes \mathbf{v}.$$

Identity: For any $\mathbf{v} \in \mathbb{R}^+$, we have $1 \otimes \mathbf{v} = \mathbf{v}^1 = \mathbf{v}$.

Vector distributivity: For any $\mathbf{u}, \mathbf{v} \in \mathbb{R}^+$ and $\lambda \in \mathbb{R}$, we have

$$\lambda \otimes (\mathbf{u} \oplus \mathbf{v}) = (\mathbf{u} \oplus \mathbf{v})^\lambda = (\mathbf{uv})^\lambda = \mathbf{u}^\lambda \mathbf{v}^\lambda = \mathbf{u}^\lambda \oplus \mathbf{v}^\lambda = (\lambda \otimes \mathbf{u}) \oplus (\lambda \otimes \mathbf{v}).$$

Scalar distributivity: For any $\mathbf{v} \in \mathbb{R}^+$ and $\lambda, \mu \in \mathbb{R}$, we have

$$(\lambda + \mu) \otimes \mathbf{v} = \mathbf{v}^{\lambda+\mu} = \mathbf{v}^\lambda \mathbf{v}^\mu = \mathbf{v}^\lambda \oplus \mathbf{v}^\mu = (\lambda \otimes \mathbf{v}) \oplus (\mu \otimes \mathbf{v}).$$

Since all the vector space axioms hold, $(\mathbb{R}^+, \mathbb{R}, \oplus, \otimes)$ is indeed a vector space.

Properties of vector spaces

Let $(V, \mathbb{F}, +, *)$ be a vector space. Then the following properties hold:

Lemma. For each $\mathbf{v} \in V$, we have that $0\mathbf{v} = \mathbf{0}$, and for each $\lambda \in \mathbb{F}$, we have that $\lambda\mathbf{0} = \mathbf{0}$.

Proof. Let $\mathbf{v} \in V$. We have

$$0\mathbf{v} = (0 + 0)\mathbf{v} \quad (\text{scalar additive identity}),$$

$$= 0\mathbf{v} + 0\mathbf{v} \quad (\text{scalar distributivity}),$$

$$\mathbf{0} = 0\mathbf{v} \quad (\text{vector additive inverses}).$$

Similarly, letting $\lambda \in \mathbb{F}$, we have

$$\lambda\mathbf{0} = \lambda(\mathbf{0} + \mathbf{0}) \quad (\text{vector additive identity}),$$

$$= \lambda\mathbf{0} + \lambda\mathbf{0} \quad (\text{vector distributivity}),$$

$$\mathbf{0} = \lambda\mathbf{0} \quad (\text{vector additive inverses}).$$

Properties of vector spaces

Lemma. For all $\mathbf{v} \in V$, we have $(-1)\mathbf{v} = -\mathbf{v}$.

Proof. Let $\mathbf{v} \in V$. Then

$$\begin{aligned}\mathbf{v} + (-1)\mathbf{v} &= 1\mathbf{v} + (-1)\mathbf{v} && \text{(scalar multiplicative identity),} \\ &= (1 + (-1))\mathbf{v} && \text{(scalar distributivity),} \\ &= 0\mathbf{v} && \text{(scalar additive inverses),} \\ &= \mathbf{0} && \text{(previous lemma).}\end{aligned}$$

So the additive inverse of $\mathbf{v}$ is $(-1)\mathbf{v}$, that is, $-\mathbf{v} = (-1)\mathbf{v}$.

Lemma. For all $\mathbf{u}, \mathbf{v} \in V$ and $\lambda \in \mathbb{F}$, if $\lambda\mathbf{u} = \lambda\mathbf{v}$ and $\lambda \neq 0$ then $\mathbf{u} = \mathbf{v}$.

Proof. Suppose $\lambda\mathbf{u} = \lambda\mathbf{v}$ and $\lambda \neq 0$. Since $\lambda \in \mathbb{F} - \{0\}$, we know that λ^{-1} exists. Pre-multiplying both sides of $\lambda\mathbf{u} = \lambda\mathbf{v}$ by λ^{-1} and applying the scalar associativity and identity axioms then implies $\mathbf{u} = \mathbf{v}$.

Lemma. For all $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$, if $\lambda\mathbf{v} = \mathbf{0}$ then either $\lambda = 0$ or $\mathbf{v} = \mathbf{0}$.

Proof. Suppose that $\lambda\mathbf{v} = \mathbf{0}$. If $\lambda = 0$, then by an earlier lemma we know $\lambda\mathbf{v} = 0\mathbf{v} = \mathbf{0}$. Otherwise, we have $\lambda \neq 0$ and $\lambda\mathbf{v} = \mathbf{0} = \lambda\mathbf{0}$, which by the previous lemma implies $\mathbf{v} = \mathbf{0}$.

Vector subspaces

Definition. If $(V, \mathbb{F}, +, *)$ is a vector space and $U \subseteq V$, then U is a **subspace** of V , written $U \leq V$, if and only if $(U, \mathbb{F}, +, *)$ is a vector space.

Example. For any vector space V , we have $\{\mathbf{0}\} \leq V$ and $V \leq V$.

Normally to confirm a set is a vector space, we have to check all eleven axioms. But when checking for a subspace, we can be more efficient.

Lemma. (**Subspace lemma**)

If $(V, \mathbb{F}, +, *)$ is a vector space and $U \subseteq V$, then $U \leq V$ if and only if:

- $U \neq \emptyset$, and
- for all $\mathbf{u}, \mathbf{v} \in U$ and $\lambda \in \mathbb{F}$, we have $\mathbf{u} + \lambda \mathbf{v} \in U$.

Proof. First notice that by setting $\lambda = 1$ in the second condition, we can confirm that U is closed under vector addition. Since U is nonempty, by picking some $\mathbf{u} \in U$ and $\lambda = -1$, we can confirm $\mathbf{u} + (-1)\mathbf{u} = \mathbf{0} \in U$, so the additive identity exists in U . Now picking $\mathbf{u} = \mathbf{0}$, we can confirm $\mathbf{0} + \lambda \mathbf{v} = \lambda \mathbf{v} \in U$ so that U is closed under scalar multiplication. Further setting $\lambda = -1$ confirms the existence of additive inverses.

The remaining vector space axioms (additive associativity and commutativity, scalar multiplicative associativity and identity, vector and scalar distributivity) are all inherited from V .

Examples of non-subspaces

Exercise. Show that $U = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x + y = 1 \right\}$ is **not** a subspace of $\mathbb{R}^2$.

Solution. U cannot be a subspace since the additive identity $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ in $\mathbb{R}^2$ is not an element of U (since $0 + 0 \neq 1$).

Exercise. Let $U = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = 0 \text{ or } y = 0 \right\}$. Show that $U \not\subseteq \mathbb{R}^2$.

Solution. U cannot be a subspace since it is not closed under addition. For example, two elements of U are $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, but $\mathbf{u} + \mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is not an element of U since $x \neq 0$ and $y \neq 0$.

Exercise. Let $U = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x \geq 0 \text{ and } y \geq 0 \right\}$. Show that $U \not\subseteq \mathbb{R}^2$.

Solution. U cannot be a subspace since it is not closed under scalar multiplication. For example, one element of U is $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and we can take the real scalar $\lambda = -1 \in \mathbb{R}$, but $\lambda \mathbf{v} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \notin U$ since $-1 \not\geq 0$.

Examples of subspaces

Exercise. Show that $U = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : x_1 - 2x_2 = 0 \right\}$ is a subspace of $\mathbb{R}^2$.

Solution. First note that U is nonempty since, for example, $\begin{pmatrix} 0 \\ 0 \end{pmatrix} \in U$.

Now let $\lambda \in \mathbb{R}$ and let $\mathbf{u}, \mathbf{v} \in U$. Then writing $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ for some $u_1, u_2, v_1, v_2 \in \mathbb{R}$, we must have that $u_1 - 2u_2 = 0$ and $v_1 - 2v_2 = 0$.

Since $\mathbf{u} + \lambda\mathbf{v} = \begin{pmatrix} u_1 + \lambda v_1 \\ u_2 + \lambda v_2 \end{pmatrix}$, we can check that

$$(u_1 + \lambda v_1) - 2(u_2 + \lambda v_2) = (u_1 - 2u_2) + \lambda(v_1 - 2v_2) = 0 + 0 = 0,$$

confirming that $\mathbf{u} + \lambda\mathbf{v} \in U$.

Thus by the subspace lemma, we have that $U \leq \mathbb{R}^2$.

Exercise. Show that $U = \{p \in \mathcal{P}_2(\mathbb{R}) : 2p(x) - xp'(x) = 0 \text{ for all } x \in \mathbb{R}\}$ is a vector space.

(Hint: Show instead that $U \leq \mathcal{P}_2(\mathbb{R})$.)

Examples of subspaces

Here are some other examples of subspaces, which you might like to try proving as an exercise:

- Geometrically, the only subspaces of $\mathbb{R}^3$ are the origin, any line through the origin, any plane through the origin, and the entire space $\mathbb{R}^3$.
- For any field $\mathbb{F}$, the set of all symmetric $n \times n$ matrices over $\mathbb{F}$ (matrices A for which $A = A^T$) is a subspace of $M_n(\mathbb{F})$.
- For any field $\mathbb{F}$ and $n \in \mathbb{N}$, we have $\mathcal{P}_n(\mathbb{F}) \leq \mathcal{P}(\mathbb{F})$.
- **Notation.** The set of all **real functions** from some nonempty set X is denoted $\mathcal{R}(X)$ (or $\mathbb{R}^X$), given by $\mathcal{R}(X) = \{f : X \rightarrow \mathbb{R}\}$. The set of all **continuous functions** from some nonempty **interval** $X \subseteq \mathbb{R}$ is denoted $\mathcal{C}(X)$. The set of all **differentiable functions** from some nonempty **interval** $X \subseteq \mathbb{R}$ is denoted $\mathcal{D}(X)$.

For any nonempty interval $X \subseteq \mathbb{R}$, we have the chain of subspaces

$$\mathcal{P}(X) \leq \mathcal{D}(X) \leq \mathcal{C}(X) \leq \mathcal{R}(X).$$